

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Questions

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA0515

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks 50

Duration :60 Minutes

Annexure A

Scenario-Based Questions

Code of Conduct:

I D.Mani Kumar Reddy(192325095) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks (100)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario	
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues	
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts	
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided	
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	

Scenario-Based Questions

1. Object Boundary Detection

In a medical image, boundaries between tissues need to be clearly identified.

- (a) Interpret the scenario and explain the importance of boundary detection.
- (b) Identify key challenges in detecting boundaries.
- (c) Apply a suitable edge detection method.
- (d) Justify your selection with logical reasoning.
- (e) Present your explanation clearly and systematically.

1. Object Boundary Detection

Answer

(a) Interpretation of the Scenario and Importance of Boundary Detection

In medical imaging, different tissues such as muscles, bones, blood vessels, and tumors often appear close together. Boundary detection is used to identify the exact borders between these tissues. Accurate boundary detection helps doctors distinguish healthy tissues from abnormal tissues and assists in diagnosis, treatment planning, and surgical procedures.

Boundary detection is one of the most important preprocessing steps in computer vision because it provides information about the shape, size, and location of objects within an image.

(b) Key Challenges in Detecting Boundaries

Several factors make boundary detection difficult:

- **Image Noise:** Medical images may contain random noise that creates false edges.
- **Low Contrast:** Adjacent tissues sometimes have very similar intensity values, making boundaries difficult to identify.
- **Blurred Edges:** Motion during image acquisition or imaging limitations can blur object boundaries.
- **Overlapping Structures:** Different tissues may overlap, making edge detection more challenging.
- **Uneven Illumination:** Variations in brightness can affect edge detection accuracy.

(c) Suitable Edge Detection Method

The **Canny Edge Detection Algorithm** is the most suitable method for this scenario.

The Canny algorithm performs the following steps:

1. Removes image noise using Gaussian filtering.
2. Calculates image gradients.
3. Applies non-maximum suppression to obtain thin edges.
4. Uses double thresholding to distinguish strong and weak edges.
5. Performs edge tracking using hysteresis to connect valid edges.

(d) Justification

The Canny Edge Detection algorithm is preferred because:

- It effectively removes noise before detecting edges.
- It produces thin and accurate boundaries.
- It minimizes false edge detection.
- It provides continuous edges even in low-contrast medical images.
- It is widely used in medical image analysis because of its high accuracy and reliability.

Compared to Sobel or Prewitt operators, the Canny algorithm produces better edge localization and fewer unwanted edges.

(e) Conclusion

Boundary detection plays a vital role in medical image analysis. Among various edge detection techniques, the Canny Edge Detection algorithm provides accurate, continuous, and reliable boundaries, making it the best choice for identifying tissues in medical images.

2. Simple Segmentation

An image contains objects clearly distinguishable from the background based on intensity.

- (a) Interpret the scenario and explain segmentation requirements.
- (b) Identify key features enabling separation.
- (c) Apply a suitable segmentation technique.
- (d) Justify your decision logically.
- (e) Present your response clearly.

2. Simple Segmentation

Answer

(a) Interpretation of the Scenario and Segmentation Requirements

Segmentation is the process of dividing an image into meaningful regions. In this scenario, the objects can be clearly distinguished from the background because they have different intensity values.

The objective is to separate the object accurately while ignoring the background.

(b) Key Features Enabling Separation

The following characteristics make segmentation easier:

- Clear difference in intensity between object and background.
- Uniform brightness within the object.
- Minimal image noise.
- High image contrast.
- Distinct object boundaries.

These characteristics allow pixels to be classified easily.

(c) Suitable Segmentation Technique

The most suitable technique is **Threshold Segmentation**.

In threshold segmentation, a threshold value **T** is selected.

- Pixels with intensity greater than **T** are classified as object pixels.
- Pixels with intensity less than **T** are classified as background pixels.

Mathematically,

$$g(x, y) = \begin{cases} 1, & f(x, y) \geq T \\ 0, & f(x, y) < T \end{cases}$$

(d) Justification

Threshold segmentation is suitable because:

- It is simple and computationally efficient.
- It provides fast segmentation.
- It works very well when object and background have distinct intensity values.
- It requires very little computation compared to advanced segmentation methods.

Since the given scenario already states that the objects are clearly distinguishable, threshold segmentation provides accurate results.

(e) Conclusion

Threshold segmentation is the most appropriate method because it efficiently separates the object from the background using intensity values. It is simple, fast, and highly accurate for images with clear contrast.

3. Complex Segmentation

An image contains multiple regions with overlapping intensity values.

- (a) Interpret the scenario and explain segmentation challenges.
- (b) Identify key issues such as intensity overlap.
- (c) Apply an appropriate advanced segmentation method.
- (d) Justify how the method handles complexity.
- (e) Provide a structured and clear explanation.

3. Complex Segmentation

Answer

(a) Interpretation of the Scenario and Segmentation Challenges

In this scenario, several objects or regions have overlapping intensity values. Since neighboring regions appear similar, simple thresholding cannot distinguish them correctly.

The segmentation algorithm must accurately separate touching and overlapping regions.

(b) Key Issues

The major challenges include:

- Overlapping intensity values.
- Touching or connected objects.
- Noise in the image.
- Weak object boundaries.
- Complex image structure.

These issues make segmentation more difficult than simple intensity-based separation.

(c) Suitable Advanced Segmentation Method

The **Watershed Segmentation Algorithm** is the most suitable technique.

The watershed algorithm treats the grayscale image as a topographic surface.

- Bright pixels represent hills.
- Dark pixels represent valleys.

- Water starts filling valleys.
- When water from different valleys meets, watershed boundaries are formed.

These boundaries separate different objects.

(d) Justification

Watershed segmentation is effective because:

- It separates touching objects.
- It accurately identifies object boundaries.
- It performs well when intensity values overlap.
- Marker-controlled watershed reduces over-segmentation.
- It is widely used in biomedical and satellite image analysis.

Compared with thresholding, watershed provides much better separation for complex images.

(e) Conclusion

Watershed segmentation is the best choice for complex images because it successfully separates overlapping objects and produces accurate segmentation even when intensity values are similar.

4. Broken Edges

After applying edge detection, object boundaries appear discontinuous.

- (a) Interpret the scenario and explain why edges are broken.
- (b) Identify key issues affecting boundary continuity.
- (c) Apply a suitable method to improve edge connectivity.
- (d) Justify your approach logically.
- (e) Present your explanation clearly.

4. Broken Edges

Answer

(a) Interpretation of the Scenario

After edge detection, some object boundaries appear broken or disconnected. These incomplete boundaries make it difficult to recognize the object's actual shape.

Continuous edges are essential for accurate object detection and image analysis.

(b) Key Issues Affecting Boundary Continuity

Broken edges occur because of:

- Image noise.
- Weak edge intensity.
- Incorrect threshold values.
- Low image contrast.
- Small gaps between edge pixels.

These problems reduce edge continuity.

(c) Suitable Method

The most suitable technique is **Morphological Closing**.

Closing consists of two operations:

1. **Dilation**

2. Erosion

Closing operation:

$$A \cdot B = (A \oplus B) \ominus B$$

where

- A = Image
- B = Structuring element

(d) Justification

Morphological Closing:

- Connects nearby edge segments.
- Fills small holes.
- Removes tiny gaps.
- Preserves the overall object shape.
- Improves boundary continuity.

Because of these advantages, closing is commonly applied after edge detection.

(e) Conclusion

Morphological Closing effectively repairs broken boundaries, resulting in continuous object edges that improve recognition, segmentation, and shape analysis.

5. Combined Requirement

An industrial inspection system requires noise removal while preserving important edges for defect detection.

- (a) Interpret the scenario and define the combined requirement.
- (b) Identify key issues involving noise and edge preservation.
- (c) Apply an appropriate sequence of techniques.
- (d) Justify the order and selection with strong reasoning.
- (e) Present your response in a clear and structured format.

5. Combined Requirement

Answer

(a) Interpretation of the Scenario

Industrial inspection systems are used to detect manufacturing defects such as scratches, cracks, holes, or missing components.

The system must remove image noise while preserving important edges because defects usually appear near object boundaries.

Both objectives must be achieved simultaneously.

(b) Key Issues

The main challenges include:

- Random image noise.
- Edge blurring caused by smoothing.
- Small defects that resemble noise.
- Maintaining accurate object boundaries.
- Achieving high inspection accuracy.

(c) Appropriate Sequence of Techniques

The recommended sequence is:

Step 1: Median Filtering

- Removes salt-and-pepper noise.
- Preserves object edges.
- Does not blur important boundaries.



Step 2: Canny Edge Detection

- Detects accurate object boundaries.
- Produces thin and continuous edges.
- Minimizes false edge detection.

(d) Justification

The sequence is important because:

- If edge detection is applied before noise removal, noise will produce many false edges.
- Median filtering first removes unwanted noise while preserving useful edge information.
- Canny edge detection then accurately detects the remaining object boundaries.
- This combination improves defect detection accuracy in industrial inspection systems.

This sequence is widely used in quality control applications.

(e) Conclusion

Applying **Median Filtering followed by Canny Edge Detection** provides the best balance between noise removal and edge preservation. This approach improves defect detection accuracy, reduces false detections, and ensures reliable industrial inspection.